

Attachment J-4

Payload Space Weight Power Cooling and Interfaces for Medium Unmanned Surface Vessel (MUSV)



March 2026

Version 1.0

1.0 Space, Weight, Power and Cooling (SWaP-C) Requirements

1.1 Interior Reservations

1.1.1 The vessel shall include a SWaP-C reservation for future Government Furnished Equipment (GFE) racks as defined as per Table 1 below:

Table 1: Interior SWaP-C Reservation for Government Furnished Equipment

Interior Payload GFE SWaP-C										
Rack Count	DESCRIPTION	WEIGHT (lbs)	HEIGHT (in) Per Unit	WIDTH (in) Per Unit	DEPTH (in) Per Unit	Front maintenance Envelope (in)	Rear maintenance Envelope (in)	HEAT DIS TO AIR BTUs/h	HVAC (CFM)	AC Power (Watts)
1	EQUIPMENT RACK	800	72	22	26	30	18	6824	500	2000
1	EQUIPMENT RACK	108	64	24	29	29	18	887	190	265
1	EQUIPMENT RACK	1200	72	23	35	26	13	5500	290	1600
1	EQUIPMENT RACK	650	61	24	41	23	18	10126	260	2967
1	EQUIPMENT RACK	671	64	23	39	30	13	4488	470	1800
1	EQUIPMENT RACK	1388	72	28	38	42	18	23549	500	6902
1	EQUIPMENT RACK	1153	71	26	44	24	18	17672	500	6510
1	EQUIPMENT RACK	800	72	22	30	30	18	9610	500	2816
1	EQUIPMENT RACK	800	72	22	30	30	18	9610	500	2816
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1.1.2 Power shall be provided in 220 V, 60 Hz, 3-ph for all racks in Table 1.

1.2 Payload Deck Requirements

1.2.1 Payload deck is capable of integrating two (2) Forty-Foot Equivalent Units (FEUs) that weigh 36.3 MT each and continuously consume 75 kW each (72.6 MT and 150 kW total).

1.2.1.1 Mounting Requirements: ISO 1161:2016 corner fittings shall be utilized as the container attachment to the payload deck and will interface with standard ISO twist-lock sockets. The payload footprints shall conform to ISO 668-2020 container dimensions. FEU payload deck locations shall also be capable of landing two TEUs within the FEU footprint.

1.2.1.2 Power Requirements: The vessel shall provide a power connection to each payload tiedown location with the following characteristics:

- a) 440 VAC, 60 Hz, 3-ph
- b) Power Usage Factor = 1.0
- c) Voltage Stability (Input Feed): $\pm 5\%$ of input voltage (440VAC)
- d) Frequency Stability (Input Feed): $\pm 0.5\text{Hz}$
- e) Harmonic Distortion (Input Feed): THD $\leq 5\%$

1.2.1.3 Payload Power Connections: Each payload location shall have a power connection capable of providing the full allocated power to each container, located no more than 10 ft from the payload location it is servicing.

1.2.1.4 Network Requirements: Each payload location shall be provided with an Ethernet connection with the other end connecting to the interior SWaP-C reservation space. If the cable is run in the weather it should be shielded and fastened to the deck.

1.2.1.5 Payload Clearance: A clearance envelope of four (4) feet shall be maintained between FEU payloads. Four (4) ft envelope shall be maintained at the lengthwise ends of the containers to allow for the opening of container doors.

1.2.1.6 Center of Gravity (CG) for Weight and Stability Analysis: the CG of each payload shall be assumed to be in accordance with Table 2. The coordinate system is depicted in Figure 1 with the origin located under the ISO fittings, along the centerline, and at the forward face of the container.

Table 2: Center of Gravity of the FEU Payload

CG (X-Direction)	CG (Y-Direction)	CG (Z-Direction)
(in)	(in)	(in)
161.98	0.79	85.52

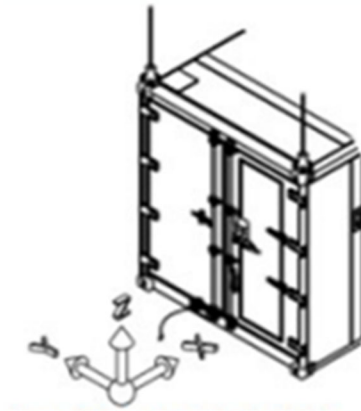


Figure 1: Coordinate System for Payload CG